

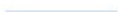
Spark and Distributed ETL

Transform analytical data from relational storage to distributed pipelines.



Learning outcomes

01  Explain partitions, shuffles, and lazy execution.

02  Translate a local transform to Spark.

03  Diagnose skew and avoid unnecessary movement.

The concept system

01

partitions

02

lazy evaluation

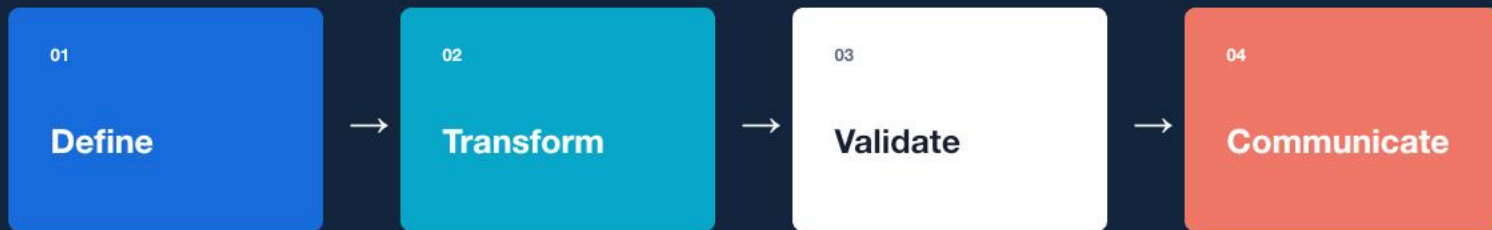
03

shuffles

04

distributed joins

A repeatable reasoning loop



Scale an event aggregation from Pandas to PySpark.

Guided lab

LAB BRIEF

Scale an event aggregation from Pandas to PySpark.

[workbook](#) / [tests](#) / [evidence](#)

01 **Frame**

Define inputs, outputs, and success criteria.

02 **Build**

Implement the smallest correct workflow.

03 **Verify**

Test normal, boundary, and failure cases.

04 **Explain**

Record the evidence and remaining limits.

Portfolio deliverable

A Spark job with execution-plan notes and data checks.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Analytical Pipeline at Two Scales

Data model 25%, SQL 25%,
Spark 25%, tests 15%,
documentation 10%.

What should remain after the lesson?

CONCEPT

partitions + lazy evaluation

PRACTICE

Scale an event aggregation from Pandas to PySpark.

EVIDENCE

A Spark job with execution-plan notes and data checks.
